

Migraine triggers

A National Migraine Centre factsheet

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Prevent migraine by learning to identify – and eliminate – common triggers

What is a migraine trigger?

A migraine trigger is any event, change, act, or any other external or internal factor that can result in migraine. Triggers might be easy to control, like avoiding dehydration, harder to control, like avoiding stress, or impossible to control, like changes in the weather!

There is no one single migraine trigger that affects everyone. A study by the National Migraine Centre found 79 per cent of patients were aware of factors that they thought triggered migraine – but most felt that several triggers at once were needed to bring about an attack.

Migraine triggers vary from one individual to the next, but there are some common triggers that affect many people with migraine.

Imagine that you have a 'migraine threshold', tolerating potential migraine triggers without an attack until a certain threshold is crossed, which might involve several factors at once. Different triggers will occur over a period of time and can build up in combination, until they cross your migraine threshold. At this point a migraine attack occurs.

Missing a meal, flickering sunlight or a lack of sleep may not always bring on an attack, but any or all of these combined with a period of stress, for example, might be enough to trigger an attack.

By identifying and addressing your triggers, you will very often be able to reduce the frequency of attacks and so use less medication.

Many patients find it possible to prevent migraines by modifying their lifestyle and avoiding triggers.

There's no definitive migraine triggers list, but here are a few of the most common.

Migraine and diet

Migraine food triggers are far less common than many people think. Certain foods, in particular cheese, chocolate, citrus fruits and dairy produce have been blamed for triggering migraine. However, the true link with migraine is uncertain since many people will crave these foods before the headache starts and may then mistakenly believe it was the food that triggered the attack.

Foods are generally not considered significant as triggers for most people. Even if foods can be shown to link to migraine, controlling other non-food triggers can make the impact of a food trigger less significant.

[Read our factsheet on migraine and food](#) to learn more, to see our recommendations for the migraine-friendly diet, and to find out about keeping a headache diary to track possible headache food [triggers](#).

through a growth spurt or involved in strenuous exercise. Many children will come home from school with a bad headache if they haven't had sufficient food often enough during the day. But insufficient food may also be an important migraine trigger in adults too.

Missing breakfast typically triggers attacks late morning, while missing lunch may trigger attacks late afternoon. If attacks are present on waking, it's worthwhile considering the time you eat your evening meal, which may be quite early. A bowl of cereal last thing at night might be all you need to prevent morning attacks.

Many people with migraine find that they need to eat frequent snacks every four hours or so during the day to avoid the peaks and troughs in blood sugar.

Sugary snacks and chocolate are fine to eat, but at the end of meal and not in place of one.

Mild dehydration

Dehydration is a common migraine trigger. You should aim to drink at least eight glasses of water per day. This is in addition to any other drinks you may have. Fizzy drinks can contain the sweetener aspartame, which some people link to their migraines.

Alcohol

Any list of triggers of migraine must include alcohol, which is probably the most frequently mentioned. Certain types of alcohol contain chemicals that can either directly affect blood vessels or provoke the release of other chemicals in the body that are thought to be involved in migraine.

Certain red wines contain more of these chemicals than drinks such as vodka or white wine.

But, whatever the drink, alcohol should be consumed in moderation. It can also be helpful to drink a glass of water alongside every alcoholic drink, to avoid the possible risk dehydration.

each day. Cola and some other soft drinks contain caffeine and should also be consumed in moderation; energy drinks, in particular, can have high levels of caffeine.

Some people find that suddenly stopping caffeine altogether can also be a trigger factor. For example, if you regularly drink caffeine-containing drinks at work but get migraines on your day off, the trigger may be caffeine withdrawal. If you suspect this, you may wish to cut down on caffeine more gradually. You should be aware that caffeine can be found in many products, including chocolate and some over-the-counter painkillers.

Change in routine

Changes in your daily routine can be behind many common migraine triggers. For example, changing sleep patterns, peaks and troughs of stress and even otherwise pleasant changes, such as a holiday, can also lead to an attack.

Sleep

The link between migraine and sleep is poorly understood. It's not uncommon to wake up with a migraine – but sleep during an attack can also resolve symptoms.

Lack of sleep is recognised as a migraine trigger, but other factors may also be important. For example, a lack of sleep can result from depression, anxiety, menopausal hot flushes or delayed bedtime due to social events, work or study – each of these could be a migraine trigger in its own right.

Many people notice that sleeping in for even just half an hour longer than usual can result in migraine. This may be one cause for weekend migraine.

If you suffer from migraine, try to keep to a fixed sleep pattern, going to bed at night and getting up in the morning at regular times. Shift workers should try to avoid frequent changes of shift times wherever possible.

Weekends

Caffeine withdrawal, caused by drinking less tea and coffee at weekends, could also be a factor. Many people who don't move around a great deal at work take unaccustomed exercise at weekends, with housework, gardening, trips to the gym and DIY projects.

Stress

Anxiety and emotion play an important role in headache and migraine.

Many people with migraine find they can cope with stress without having a migraine at the time, but will have attacks when they relax. It's not uncommon for migraine to hit following a work deadline, after a big event like a wedding or on the first day of a holiday.

However, stress rarely occurs without a knock-on effect for other possible triggers too, such as missed meals, poor sleep and increased muscle tension.

Although stress is often unavoidable, it's important to try to reduce the effects of other avoidable triggers by eating regularly and getting adequate sleep. This can also help you cope with the stress better.

Travel

How often have you found that a long journey by car or plane results in a migraine? Travel is associated with a host of potential migraine triggers: lack of sleep from preparation for the trip and from the trip itself, stress, missed or delayed meals, noise, and dehydration.

It's unclear if pressure changes in aircraft triggers migraine, particularly with improved cabin pressures in most modern planes.

Weather

Changes in pressure are often suggested as a trigger for migraine, although there's no clear evidence. In the UK, a study in London found no evidence that the weather affects

Computer work is often a cause of headache. The causes could be both related to the flickering screen and to how you sit and work at the computer.

You can minimise the risks by:

- using an anti-glare screen
- positioning your monitor in daylight, rather than under fluorescent lighting
- changing the refresh rate of your computer monitor
- checking whether you may need glasses for close-up work and reading
- taking a break every half an hour, gently stretch your head, neck and arms
- regularly rest your eyes on something far in the distance and blinking hard several times

Exercise

If you're unfit, strenuous exercise can trigger an attack, as well as causing muscle aches and pains.

This puts many people off taking exercise, but regular exercise can actually help prevent migraine.

A new exercise programme should start off gently, building up the pace gradually over several weeks.

It's important to keep the exercise sessions regular. Short, frequent sessions are more beneficial than long, infrequent sessions.

As you get fitter, you'll have an improved blood sugar balance and better breathing. You will also have better pain control, with exercise stimulating the body to release natural pain- controlling endorphins and enkephalins, relieving depression, and promoting a general sense of well-being.

Hormones

In a study undertaken at the National Migraine Centre , more than 50 per cent of women reported that they were more likely to have a migraine attack around the time of their menstrual period.

Although most women have attacks at other times of the cycle too, a small percentage of women have attacks that are exclusively associated with menstruation.

Other hormonal changes such as use of hormonal contraception can exacerbate migraine symptoms for some women and improve them for others.

The years leading up to the menopause are typically associated with increased frequency of migraine, particularly menstrual migraine. The hormonal fluctuations associated with worsening of migraine at this time can be controlled by using hormone replacement therapy, which may be particularly useful if other menopausal symptoms such as flushes and sweats are present.

Find out more with our factsheets on [Menstrual migraine](#) and [Migraine and contraception](#). Always speak to GP or a headache specialist before beginning treatment.

Illness

Most people will get a headache when they have a cold or a viral infection, but migraine can also be triggered. It's not clear if the illness is a trigger in its own right or if it lowers the migraine threshold so that fewer triggers will lead to an attack.

If you're coming down with a cold, stock up on migraine treatments as well as cold remedies – but make sure that you don't overdose on painkillers.

Neck and back pain

Physical causes such as these require physical treatments, although it may be several months before you see any benefit. Sometimes, over-enthusiastic osteopathy or physiotherapy can trigger attacks rather than help. Like exercise, a gentle start is the key to long-term benefit.

Jaw joint dysfunction

If you find that your jaw clicks when you eat or even locks, or you frequently wake with migraine after grinding your teeth at night, you may have a problem with your jaw joint.

Pain and tenderness in the jaw joints can be associated with tension in the muscles controlling the jaw. This may lead to headache, often daily, but can also trigger migraine.

How do I find my migraine triggers?

With all the potential headache triggers, it can be tough to identify those to which you are particularly susceptible.

Everyone with migraine should start by keeping a simple headache diary, which can be so helpful in reducing the impact of attacks. You can download our [headache diary template here](#).

But when you're identifying triggers, you should also keep a separate trigger diary too. You can just use a notepad, with a day per page. Write down every time you encountered a suspected trigger – perhaps one of the triggers listed above or something you suspect to be linked to your attacks.

So, make sure to include if you skipped a meal, if you worked late, if you spent a bit longer in bed, if you had a glass of wine – anything and everything that might be a potential trigger.

Add your triggers every day, just before you go to bed. It's important not to wait until you have had attack and then try to remember all the possible triggers from the past few days – you're unlikely to remember them all.

How do I make sense of my migraine trigger diary?

It's common to hear the question, 'What triggers a migraine attack?'. But a more useful question is often, 'How many triggers are needed to begin an attack?'.

Even your usual daily routine could include some triggers that you're not aware of if you generally remain below the migraine threshold until faced with a few extra triggers.

Compare your trigger diary with your [headache diary](#). See if there was a build up of triggers in the days before each attack. Does a pattern emerge, with certain possible triggers cropping up several times?

You should be able to divide your list of triggers into two groups – those that you can do something about (such as missing meals, drinking red wine) and those that are out of your control (like the menstrual cycle or travelling).

First, try to deal with the triggers that you have some control over. Cut out suspected triggers one at a time – if you try to deal with them all at once you will not know which are most relevant to you.

Try to compensate, so that if you are having a particularly stressful time, take care to eat regularly and find ways to unwind before you go to bed. Just because you're facing one trigger doesn't mean it has to lead to a migraine, particularly if you can reduce the impact of the other possible triggers.

If your attacks regularly start late morning or late afternoon, look at your mealtimes. A mid-morning or mid-afternoon snack may be all that is needed to prevent attacks. Similarly, if you have an early evening meal and wake with an attack, try a snack before you go to bed.

Can identifying triggers really make a difference?

Yes! It can take time, effort and motivation to record and deal with migraine triggers, but studies show it's possible to halve your number of attacks this way – which could make

triggers.

Book an appointment with the experts: a consultation with a leading headache specialist through the National Migraine Centre could help you manage your triggers. **Book your consultation now.**

Speak to a leading GP headache specialist or consultant neurologist remotely, from the comfort of your home.

The National Migraine Centre has helped thousands of people like you to take control of headache. Get expert advice with specialist consultations, access the latest treatments and anti-CGRP medications, and book procedures such as Botox and nerve block.

Get back to living: book a consultation today

Book a consultation

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